

SMART BORDER INTRUSION DETECTION
NETWORK...

DEPARTMENT: Electronics and Communication Engineering

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ABSTRACT:

This project presents a Smart Border Intrusion Detection Network using embedded systems and IoT technology. Traditional border surveillance systems are resource-intensive and prone to human error. To overcome these challenges, a low-cost and efficient system is developed using ESP32 microcontroller, PIR sensor, and vibration sensor. The system detects motion and disturbances in real-time and triggers alerts using buzzer and LED indicators. Dual sensor fusion improves detection accuracy and reduces false alarms. The system is tested under various conditions and shows reliable performance, making it suitable for large-scale deployment.

INTRODUCTION:

1. What is IOT ?

Internet of Things (IoT) refers to interconnected devices that collect and exchange data using the internet.

2. Problem Statement

Traditional border monitoring systems require continuous human surveillance, which is costly, inefficient, and prone to errors. Existing systems are also not scalable for long-distance borders.

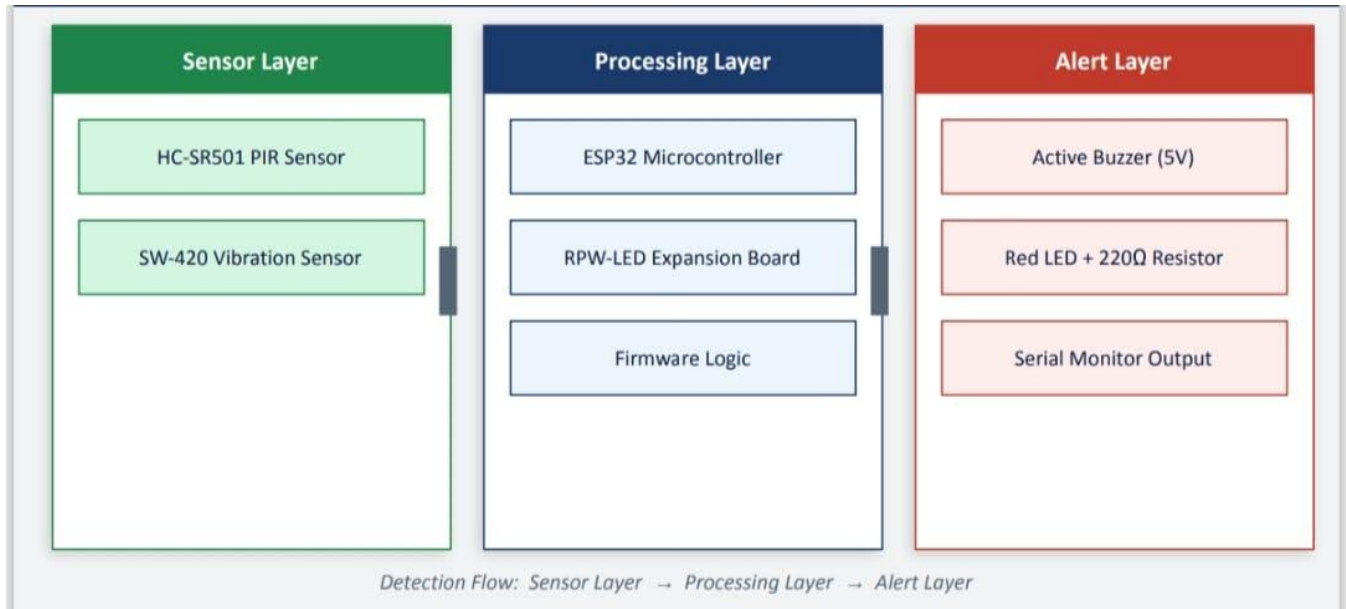
3. Objective of the Project

- To design a low-cost intrusion detection system
- To detect motion and vibration using sensors
- To provide real-time alert system
- To reduce false alarms using dual-sensor fusion

LITERATURE SURVEY:

- Existing systems use radar and thermal imaging, which are expensive
- Manual monitoring systems are prone to fatigue and errors
- IoT-based systems provide cost-effective and scalable solutions
- Sensor-based intrusion detection improves accuracy
- Embedded systems enable real-time processing

SYSTEM ARCHITECTURE:



COMPONENTS REQUIRED:

Hardware

- ESP32 DevKit v1
- PIR Sensor (HC-SR501)
- Vibration Sensor (SW-420)
- Active Buzzer
- Red LED + 220Ω resistor
- I2C LCD Display

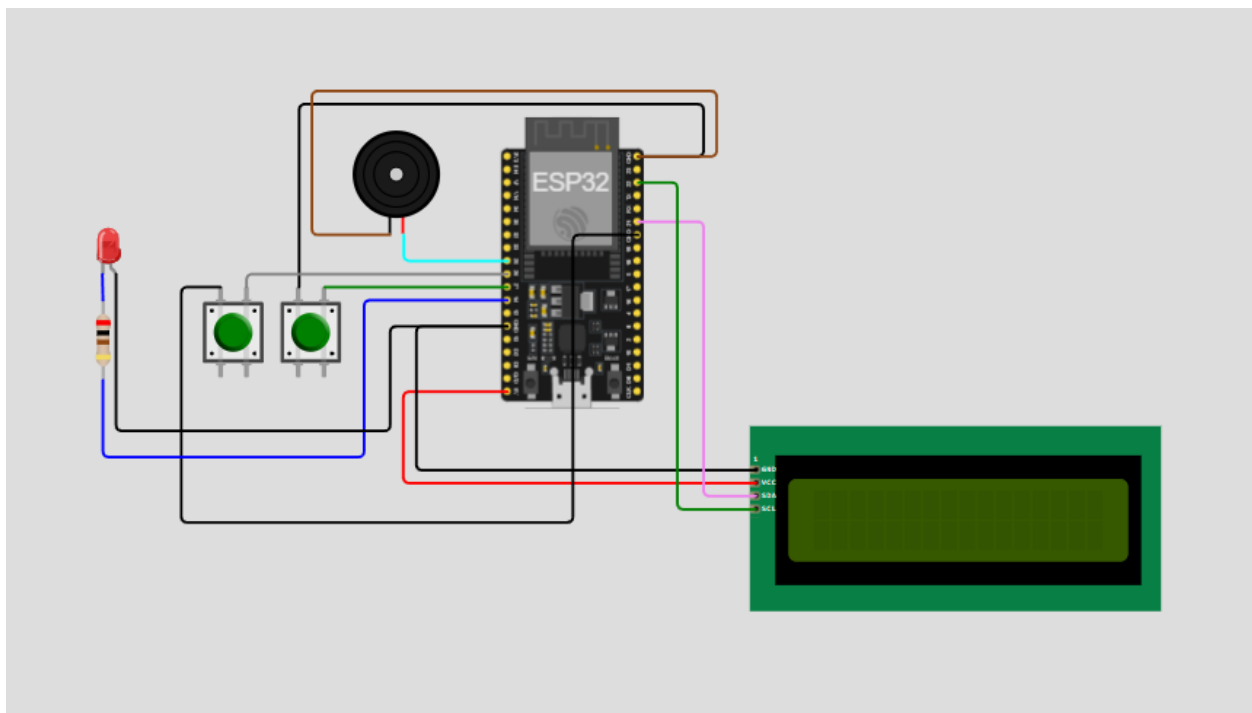
Software

- Arduino IDE
- Embedded C / Arduino Programming

WORKING PRINCIPLE:

1. System powers ON and initializes GPIO pins
2. ESP32 connects to system
3. Sensors continuously monitor environment
4. If motion or vibration detected → signal sent to ESP32
5. System checks cooldown condition
6. Alert is triggered:
 - Buzzer sound
 - LED indication
7. Cooldown timer prevents repeated alerts

CIRCUIT DIAGRAM:



SOFTWARE / CODE:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#define PIR 27
#define VIB 26
#define LED 14
#define BUZZER 25
LiquidCrystal_I2C lcd(0x27, 16, 2);
unsigned long lastTrigger = 0;
const int cooldown = 3000;
void setup() {
    Serial.begin(115200);
    pinMode(PIR, INPUT_PULLUP);
    pinMode(VIB, INPUT_PULLUP);
    pinMode(LED, OUTPUT);
    pinMode(BUZZER, OUTPUT);
    lcd.init();
    lcd.backlight();
    lcd.print("System Ready");
    delay(1500);
    lcd.clear();
}
void loop() {
    bool motion = digitalRead(PIR) == LOW;
    bool vibration = digitalRead(VIB) == LOW;
    if ((motion || vibration) && millis() - lastTrigger > cooldown) {
        lastTrigger = millis();
        digitalWrite(LED, HIGH);
        digitalWrite(BUZZER, HIGH);
        lcd.clear();
        if (motion && vibration) {
            lcd.print("HIGH ALERT!");
            lcd.setCursor(0,1);
            lcd.print("MOTION+VIB");
        }
        else if (motion) {
            lcd.print("MOTION DETECT");
        }
        else {
            lcd.print("VIBRATION!");
        }
        delay(800);
        digitalWrite(BUZZER, LOW);
    }
    if (!motion && !vibration) {
        digitalWrite(LED, LOW);
        lcd.setCursor(0,0);
        lcd.print("NO INTRUSION ");
        lcd.setCursor(0,1);
        lcd.print("SYSTEM SAFE ");
    }
    delay(100);
}
```

RESULTS:

1.PIR Sensor Performance

- Detects motion up to 3.5 – 4 meters (calibrated)
- Works reliably for human movement detection

2. Vibration Sensor Performance

- Successfully detects disturbances and vibrations
- Triggers alert immediately when activated

3.Response Time

- Buzzer response time: less than 50 ms
- LED response time: less than 50 ms

4.Alert System

- Audio (buzzer) and visual (LED) alerts work properly
- Different alert patterns based on sensor input

5.Cooldown Mechanism

- 15-second delay prevents repeated alerts
- Reduces false triggering effectively

6.Dual Sensor Fusion

- Improves accuracy of detection
- Provides HIGH alert when both sensors are triggered

7.Serial Monitor Output

- Sensor readings are displayed correctly
- Helps in monitoring and debugging

8.Overall System Performance

- All test cases passed successfully
- System is stable, reliable, and ready for real-time use

APPLICATIONS:

- 1) Border security systems
- 2) Military surveillance
- 3) Smart home security
- 4) Industrial monitoring
- 5) Restricted area protection

CONCLUSION:

The project successfully demonstrates a low-cost and efficient intrusion detection system using IoT and embedded systems. The use of dual sensors improves detection accuracy and reduces false alarms. The system is reliable, scalable, and suitable for real-world security applications. Future improvements can include cloud integration and mobile alerts.

OUTPUT:

